



SHIVNETRA 47

SHIV-NETRA47 AIR DEFENCE SYSTEM PRIVATE LIMITED
AUTONOMOUS AIR SYSTEMS • SWARM INTELLIGENCE • TACTICAL LOGISTICS

🇮🇳 MAKE IN INDIA • ATMANIRBHAR BHARAT

Engineering Collaboration: BCN In-House Technology Team

PROGRAM BLUEPRINT & TABLE OF CONTENTS

CAPABILITY ARCHITECTURE
& TECHNICAL INDEX

A comprehensive operational reference detailing sovereign autonomous swarm intelligence, heavy-cargo hybrid VTOL logistics, and frontline border defence architectures engineered to Indian Armed Forces requirements.

PART I TACTICAL UAS

SWARM & OVERWATCH ISTAR

AUTONOMOUS ISR & CONTESTED AIRSPACE

01 EXECUTIVE VISION & DOCTRINE PAGE 03

High-altitude border operational requirements and multi-tier sovereign air defence integration.

02 HETEROGENEOUS SWARM C2 PAGE 04

Distributed peer-to-peer RF mesh routing, decentralized patrol logic, and dynamic collision avoidance.

03 NANO-SCOUTS & OVERWATCH RELAYS PAGE 05

Micro air vehicle payloads (350g) teamed with 20kg endurance relays for multi-spectral EO/IR surveillance.

04 SORA SAFETY & DGCA COMPLIANCE PAGE 06

Specific Operations Risk Assessment (SAIL II/III), emergency FTS mitigation, and military airspace clearance.

PART II CARGO VTOL

TACTICAL LOGISTICS VTOL

RUNWAY-FREE FORWARD RESUPPLY

05 HYBRID AIRFRAME ARCHITECTURE PAGE 07

Vertical lift rotor transition to high-efficiency fixed-wing cruise for high-altitude mountain posts.

06 AMMUNITION & SUPPLY TRANSIT PAGE 08

Forward line resupply handling 5–10kg to 200kg payloads across 100+ km radius in zero-infrastructure terrain.

07 TRAUMA CARE & PRECISION WINCH PAGE 09

Blood plasma cold-chain pods, autonomous tethered winch delivery, and emergency parachute recovery.

08 HEAVY-LIFT HYBRID POWERTRAINS PAGE 10

Multi-fuel electric hybrid propulsion, sub-zero battery thermal conditioning, and 180-min cruise endurance.

PART III MAKE IN INDIA

SOVEREIGN INFRASTRUCTURE

PRODUCTION, C2 & PROCUREMENT

09 UNIFIED TACTICAL GCS CONSOLE PAGE 11

Ruggedized common command console controlling simultaneous swarm reconnaissance and cargo VTOL sorties.

10 EW & NAVIC TELEMETRY HARDENING PAGE 12

Anti-jamming frequency-hopping RF, sovereign NAVIC satellite synchronization, and optical dead-reckoning INS.

11 INDIGENOUS MANUFACTURING PLAN PAGE 13

Sovereign carbon-composite fabrication, localized component assembly, and IDEX / MoD procurement roadmap.

12 CORPORATE & DEFENCE CONTACT PAGE 14

Company legal credentials, technology collaboration frameworks, field demonstration schedule, and contact.

SOVEREIGN DOCTRINE

UNIFIED AIR ARCHITECTURE

One tactical ecosystem linking micro swarm scouts, overwatch relays, and heavy cargo VTOLs.

ALL-TERRAIN READINESS

HIMALAYAN ALTITUDE RATED

Runway-free vertical departure and flight envelope certified up to 8,000–16,000 ft AMSL.

CONTESTED SPECTRUM

NAVIC + MESH ENCRYPTION

Resilient anti-jam frequency hopping with autonomous fallback under electronic warfare denial.

MAKE IN INDIA

100% INDIGENOUS IPR

Domestic manufacturing base aligned with Ministry of Defence procurement guidelines.

MULTI-DOMAIN UNMANNED AERIAL ECOSYSTEM

Standardized Indian Defence Architecture • Modular Payloads • High-Altitude Field Trials Roadmap

● 12 CORE TECHNICAL SECTIONS

▲ 3 OPERATIONAL PILLARS

■ SOVEREIGN IPR